

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460300.3198 +/- 0.0054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.053 +/- 0.049
Transit depth (Rp/Rs)^2	0.28 +/- 0.52 [%]
Orbital Inclination (inc)	79.06 +/- 2.88
Airmass coefficient 1 (a1)	0.0408 +/- 0.0027
Airmass coefficient 2 (a2)	1.43 +/- 0.41
Scatter in the residuals of the lightcurve fit is	6.11 %
Best Comparison Star	#3 - [562, 26]
Optimal Aperture	23.05
Optimal Annulus	93.87
Transit Duration (day)	0.00016 +/- 0.00092

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.053 ± 0.049 vs 0.1368 (deviation 1.71σ)
Duration fit vs published	0.00016 d vs 0.0758 d (deviation 82.25σ)
Mid-time O–C	-2202.4 min
Depth SNR	1.1
Residual scatter	6.11 %

Light curve



Figure 1 — FinalLightCurve_WASP-36 b_2023-12-22.png (SHA-256 841d0dc5b6702dd4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [557, 79], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d10a00a22f653ab1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), WASP-36231223062415.FITS → WASP-36231223101216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-231223.log (dc6986db1a14...), FinalLightCurve_WASP-36 b_2023-12-22.png (841d0dc5b670...), FinalLightCurve_WASP-36 b_2023-12-22.csv (3724bf9011d6...)

Compute resources

Wall time 541.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 14:01Z.